

PULMONARY CIRCULATION

V E R S U S

SYSTEMIC CIRCULATION

Pulmonary circulation carries deoxygenated blood from the right ventricle of the heart to the lungs through the pulmonary artery

Carries oxygenated blood from the lungs to the left atrium of the heart by the pulmonary vein

Composed of pulmonary artery and pulmonary vein

Carries blood to the lungs

Helps to release carbon dioxide from the blood while dissolving oxygen in the blood

Systemic circulation carries oxygenated blood from the left ventricle of the heart to the rest of the body by the aorta

Carries deoxygenated blood from the body to the right atrium of the heart by the superior and inferior vena cava

Composed of inferior and superior vena cava, aorta, and other small blood vessels

Carries blood throughout the body

Helps to provide nutrients and oxygen to the metabolizing cells in the body

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Differences between Systemic and Pulmonary Circulations	
Systemic Circulation	Pulmonary Circulation
1. It is a larger circuit, which supplies blood to the various parts of the body and back to the heart.	1. It is smaller circuit which carries blood to the lungs and back to the heart.
2. It supplies oxygenated blood to all parts of the body.	2. It carries deoxygenated blood to the lungs for oxygenation.
3. The blood is pumped by left ventricle and received by the right atrium.	3. The blood is pumped by right ventricle and received by left atrium.

SYMPATHETIC NERVOUS SYSTEM VERSUS PARASYMPATHETIC NERVOUS SYSTEM

Sympathetic nervous system is a part of the autonomic nervous system that serves to accelerate the heart rate, constrict blood vessels, and raise blood pressure	Parasympathetic nervous system is a part of the autonomic nervous system that serves to slow the heart rate, increase intestinal and glandular activity, and relax the sphincter muscles
Originates from cranial, thoracic, and lumbar regions of the central nervous system	Originates from cranial and sacral regions of the central nervous system
Prepares the body for an intense physiological activity	Relaxes the body by inhibiting high energy functions
Action is a quick response	Action is a slow response
Ganglions are found close to the central nervous system.	Ganglions are found away from the central nervous system but close to the effector
Pre-ganglionic fibers are short	Pre-ganglionic fibers are long
Post-ganglionic fibers are long	Post-ganglionic fibers are short
A large number of post-ganglionic fibers are found	A small number of post-ganglionic fibers are found
Covers a large area in the body	Covers a small area in the body
Generates a diffused effect at its target area	Generates a localized effect at its target area
Noradrenaline is released at the effector	Acetylcholine is released at the effector
Generates an excitatory homeostatic effect	Generates an inhibitory homeostatic effect
Increases heart beat, blood level, and metabolic rate	Decreases heart beat, blood level, and metabolic rate
Raises the sensory awareness	Restores sensory awareness to the normal level
Dilates the pupil of the eye	Stimulates the pupil of the eye
Inhibits the saliva secretion	Stimulate the secretion of saliva
Dilates the bronchial tubules	Constricts the bronchial tubules
Releases adrenaline from adrenaline glands	Have no action on the adrenaline gland
Increases the activity of the digestive system	Decreases the activity of the digestive system
Increases the rate of glycogen breakdown	Has no effect on the glycogen breakdown
Decreases the urinary output and contracts the rectum	Increases the urinary output and relaxes the rectum

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TRACHEA

VERSUS

BRONCHI

TRACHEA

A large membranous tube reinforced by rings of cartilage, extending from the larynx to the bronchial tubes and conveying air to and from the lungs

Airways which connect the larynx to the bronchi

There is only one trachea

Wall is thin

Contains C-shaped cartilaginous rings

BRONCHI

Any of the major air passages of the lungs which diverge from the windpipe

Airways which connect the trachea to the lungs

There are two bronchi, which branch further

Thick walls due to increasing amounts of smooth muscles

Contain irregularly arranged plates and islands

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Differences between Cerebrum and Cerebellum

Cerebrum

- Part.** It is a part of fore brain.
- Size.** Cerebrum constitutes 80% of brain.
- Position.** It forms the front, superior and lateral sides of the brain.
- Components.** Cerebrum is made of two parts called cerebral hemispheres.
- Cavities.** It contains two cavities called lateral ventricles.
- Seat.** It is seat of intelligence and memory.
- Control.** Cerebrum controls intelligence, ~~movements~~ speech, sight, smell, taste, hearing and other sensations.

Cerebellum

- It is part of hind brain.
- It constitutes 12-5% of brain.
- It lies in the posterior region of brain.
- Cerebellum has three parts, two lateral cerebellar hemispheres and one central vermis.
- A cavity is nearly absent.
- It coordinates muscular activity.
- Cerebellum maintains equilibrium of the body.

ns.

Cerebrum	Cerebellum
It is the largest part of the forebrain.	It is a smaller part of the hind brain.
It is the seat of memory, will power, consciousness, intelligence.	It coordinates muscular activities and controls posture and balance of the body.
It has many convolutions and has ventricles.	Convolutions and ventricles are absent.
It is divided into four major lobes – parietal, frontal, temporal and occipital lobe.	No lobes are present in cerebellum.
It forms the front, lateral and superior sides of the brain.	It lies in the posterior region of the brain.

URETER

V E R S U S

URETHRA

Ureter refers to a duct through which urine passes from the kidney to the bladder

Drains urine from the kidney to the urinary bladder

There are two ureters in the urinary system

In the middle part of the urinary system

A thin and long tube

Considered as a part of the urinary system in males

Urethra refers to a duct through which urine passes from the bladder to the outside of the body

Drains urine from the bladder to the outside of the body

There is only a single urethra in the urinary system

In the last part of the urinary system

A broad and short tube

Considered as a part of both urinary and reproductive system

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	Ureter		Urethra
1	It is a tube that leads out of each kidney.	1	It is a canal-like opening at the bottom of the urinary bladder.
2	These tubes carry urine from the kidneys of the urinary bladder.	2	This opening excretes urine out of the body.
3	There are two ureters.	3	There is only one urethra.

CENTRAL NERVOUS SYSTEM VERSUS PERIPHERAL NERVOUS SYSTEM

Central nervous system consists of the brain and spinal cord

Nerve axons consist of slender projections and carry significantly short nerve impulse

Major function is to organize and analyze the information obtained from sensory organs

A damage causes a global effect on the body

Most of the nerves are incapable of regenerating its nerve fibers

Peripheral nervous system consists of sensory receptors, sensory neurons, and motor neurons

Composed of long nerve fibers with a length up to 1m

Major function is to transmit sensory information to the central nervous system and pass out motor impulses to the effector organs

A damage causes a local effect on the body

Most of the nerves can be regenerated

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Central vs Peripheral Nervous System

More Information Online WWW.DIFFERENCEBETWEEN.COM

Central Nervous System

Peripheral Nervous System

DEFINITION

Central nervous system is the central processing unit of the nervous system of our body.

Peripheral nervous system is the connection line between the central nervous system and the overall body parts.

COMPOSED OF

Brain and the spinal cord

Somatic and autonomic nervous system.

PROTECTED BY

Skull

Vertebral column

FUNCTION

Storing, comprehending and executing information in our body.

Transmission information from and to the central nervous system from the body parts.

DAMAGE

Can lead to global damage.

Can lead to localized damage.

SMALL INTESTINE VERSUS LARGE INTESTINE

Small intestine is a part of the intestine, which runs between the stomach and the large intestine	Large intestine is the terminal part of the intestine, which is wider and shorter than the small intestine
Long	Shorter
Narrow	Broader
Duodenum, jejunum, and ileum are the three components	Cecum, colon, rectum, and anal canal are the four components
Consists of circular folds in the internal surface	Lacks circular folds on the internal surface
Consists of villi in the internal surface	Do not consist of villi
Some parts contain Peyer's patches	Does not contain Peyer's patches
Contains circular layers of longitudinal muscles	Longitudinal muscles are arranged in three bands called Teniae coli
Does not contain haustra	Contains haustra
Lacks epiploic appendages	Contains epiploic appendages
Wall is smooth	Wall is sacculated
Involved in digestion	Digestion is not a function
Secretes hormones	Does not secrete hormones
Shows small movements	A comparatively fixed structure
Absorbs most of the nutrients in the digested food	Absorbs water from the indigestible portion of the food

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CRANIAL NERVES

V E R S U S

SPINAL NERVES

Cranial nerves are the nerves that arise directly from the brain and pass through separate apertures in the skull

Comprise 12 nerve pairs

Numbered I to XII

Distributed in the head, neck and facial regions

May contain sensory/motor/mixed neurons

Involved in vision, sense of the smell, hearing, sense of taste, and eye movements

Form dorsal and ventral roots

Spinal nerves are a series of paired nerves that originate from the nerve roots of the spinal cord on both sides

Comprise 31 nerve pairs

Classified into five groups as 8 cervical nerve pairs, 12 thoracic nerve pairs, 5 lumbar nerve pairs, 5 sacral nerve pair, and 1 pair of coccygeal nerve pair

Distributed in the skin, sweat glands, mucosa, blood vessels, joints, and skeletal muscles

Composed of both sensory and motor neurons

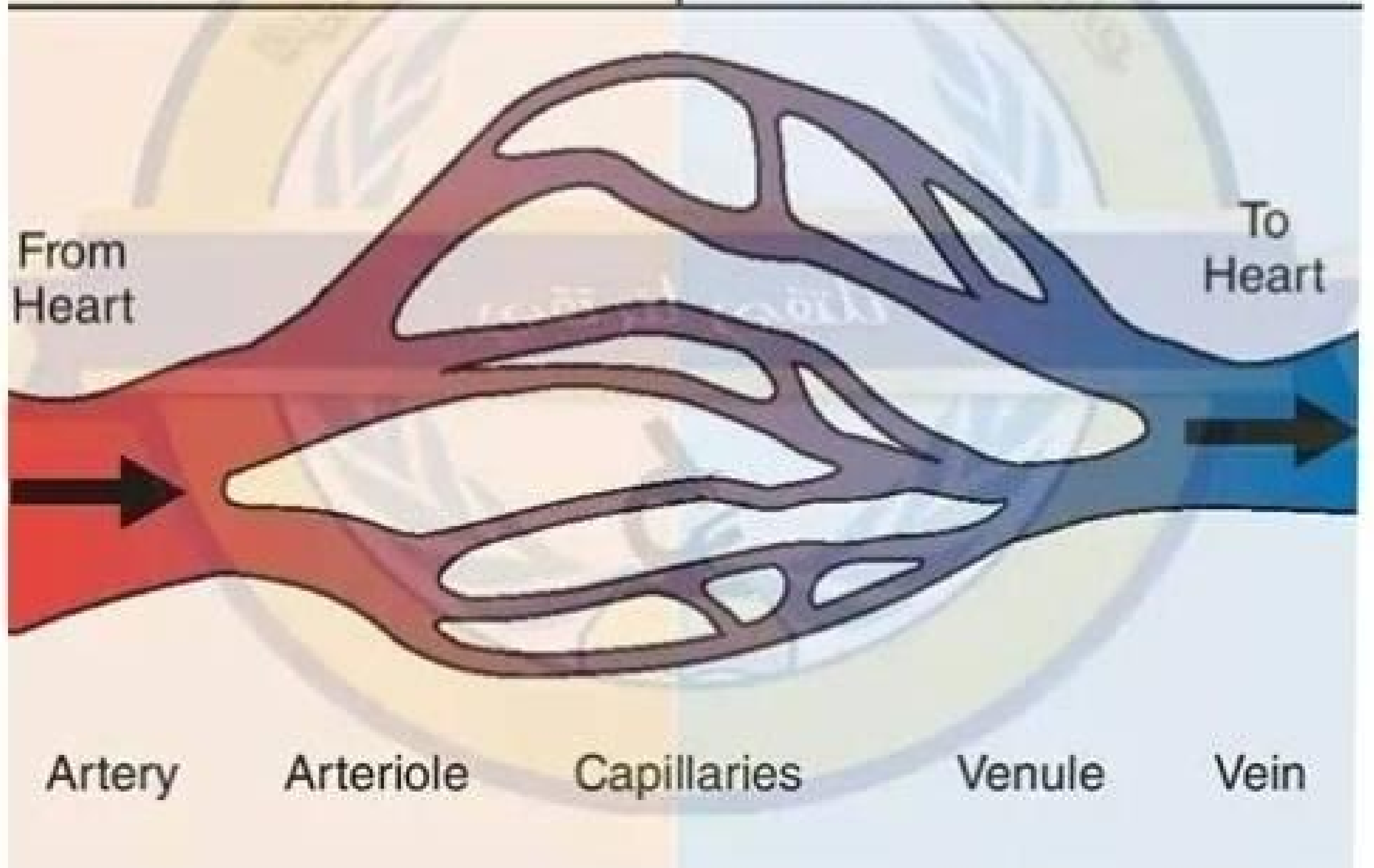
Involved in movement, sensation, and sweat secretion

Do not form dorsal and ventral roots

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ARTERIES VERSUS VEINS

Arteries	Veins
1. Carry blood from the heart, carry oxygenated blood (except pulmonary artery) 2. Normally bright red in color 3. Elastic walls that expand with surge of blood 4. No valves 5. Can feel a pulse	1. Carry blood to the heart, carry deoxygenated blood (except pulmonary vein) 2. Normally dark red in color 3. Thin walls/less elastic 4. Valves 5. No pulse



Veins	Arteries
1) Moves towards the heart.	1) Moves away from the heart.
2) Collects blood from body organs.	2) Distributes blood to the body organs.
3) Blood pressure in veins is low.	3) Blood pressure is high in arteries.
4) Valves are present.	4) Valves are absent.
5) Carry deoxygenated blood, except pulmonary vein.	5) Carry oxygenated blood except pulmonary artery.
6) Veins start in blood capillaries.	6) Arteries end in capillaries.
7) They can be seen subcutaneously.	7) They are deep seated.
8) They collapse when there is no blood in it or cut across.	8) They do not collapse when there is no blood in it.
9) Veins are further divided into venules.	9) Arteries are further divided into arterioles.
10) Veins are usually flattened or collapsed with thin walls.	10) They are round and relatively thick walled.
11) Veins have large lumen.	11) Arteries have small lumen.
12) Veins are bluish in colour.	12) Arteries are reddish in colour.
13) Veins show sluggish movement of blood.	13) Arteries show spurty movement of blood giving pulse.
14) If venous wall is injured, blood comes out, collects in a pool in a small area around vein.	14) If arterial wall is injured, the blood comes out like a fountain in a large area all around the artery.

INVERSION AND EVERSION

Inversion - movement of the sole of the foot toward the body's midline

Example: twisting the ankle in



Eversion – movement of the sole of the foot away from the body's midline

Example: twisting the ankle out



INVERSION & EVERSION

- **Inversion** is the movement of the sole towards the median plane (as when an ankle is twisted).

- Eversion the anatomical term of motion denoting the movement of the sole of the foot away from the median plane

Inversion

Eversion

